

Math 211 First Exam

11 March 2018

1. [5 pts] If $f(x, y) = x^2y^2 - xy^3 - x^2 + xy$, which of the following is not a critical point of f ?
- (A) $(0, 1)$ (B) $(0, -1)$ (C) $(1, 1)$ (D) $(1, -1)$ (E) $(0, 0)$
2. [5 pts] The tangent line to the curve $\mathbf{r}(t) = t^{-1}\mathbf{i} + (t-1)\mathbf{j} + t^2\mathbf{k}$ at the point $(1, 0, 1)$ contains the point
- (A) $(-1, 2, 5)$ (B) $(-2, 3, 5)$ (C) $(3, -3, -2)$ (D) $(\frac{1}{2}, 1, 4)$ (E) $(1, 2, 2)$
3. [5 pts] The limit of $f(x, y) = \frac{2x+y}{3x+2y}$ as $(x, y) \rightarrow (0, 0)$ is
- (A) $\frac{3}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) 0 (E) undefined
4. [5 pts] The gradient of $f(x, y) = \sqrt{x^2 + xy + y^2}$ at the point $(3, -2)$ is
- (A) $\frac{3}{2\sqrt{19}}$ (C) $\frac{2}{\sqrt{7}}\mathbf{i} - \frac{1}{2\sqrt{7}}\mathbf{j}$ (E) $\frac{1}{\sqrt{19}}\mathbf{i} + \frac{7}{2\sqrt{19}}\mathbf{j}$
(B) $-\frac{1}{\sqrt{10}}\mathbf{i} + \frac{\sqrt{5}}{2\sqrt{3}}\mathbf{j}$ (D) $-\frac{4}{\sqrt{7}}\mathbf{i} + \frac{3}{\sqrt{7}}\mathbf{j}$
5. [5 pts] If z is a function of x and y defined implicitly by the equation $xy^2z^3 - x^3yz^2 + 6 = 0$, then the value of $\frac{\partial z}{\partial x}$ at the point $(1, 2, -1)$ equals
- (A) $\frac{3}{2}$ (B) $-\frac{7}{4}$ (C) $\frac{2}{5}$ (D) $-\frac{11}{16}$ (E) $\frac{5}{8}$
6. [15 pts] Find the curvature of the curve $y = \ln(1 - e^{-x})$ at the point $(\ln 2, -\ln 2)$.
7. [15 pts] Use definition to find the directional derivative of $f(x, y) = x^2 - y^2$ at the point $(2, 1)$ in the direction of the vector $\mathbf{a} = \mathbf{i} - \mathbf{j}$.
8. [15 pts] Let $w = f(x, y)$ where $x = e^r \cos \theta$, $y = e^r \sin \theta$, and f is differentiable. Show that
- $$\left(\frac{\partial w}{\partial r}\right)^2 + \left(\frac{\partial w}{\partial \theta}\right)^2 = e^{2r} \left[\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 \right].$$
9. [15 pts] Let $f(x, y, z) = (y + z)e^{x+z}$.
- (a) Find an equation for the tangent plane at the point $(-1, 2, 1)$ to the surface $f(x, y, z) = 3$.
- (b) Use a linear approximation to estimate $f(-1.03, 2.01, 0.98)$.
10. [15 pts] Find the local extrema and saddle points, if any, of the function

$$f(x, y) = x^2 + 2y^2 - x^2y + 2.$$

SOLUTIONS

1. The answer is (D).

This is obtained by evaluating at each given point the partial derivatives

$$f_x = 2xy^2 - y^3 - 2x + y \quad \text{and} \quad f_y = 2x^2y - 3xy^2 + x.$$

2. The answer is (A).

We have $\mathbf{r}'(t) = -t^{-2}\mathbf{i} + \mathbf{j} + 2t\mathbf{k} \implies \mathbf{r}'(1) = \langle -1, 1, 2 \rangle$.

Hence a vector equation for the tangent line is $\langle x, y, z \rangle = \langle 1 - t, t, 1 + 2t \rangle$.

The point $(-1, 2, 5)$ satisfies these equations with $t = 2$.

3. The answer is (E).

$\lim_{x \rightarrow 0} f(x, 0) = \frac{2}{3}$ and $\lim_{y \rightarrow 0} f(0, y) = \frac{1}{2}$. Thus the given limit does not exist by the two path rule.

4. The answer is (C).

$$\nabla f(x, y) = f_x \mathbf{i} + f_y \mathbf{j} = \frac{2x + y}{2\sqrt{x^2 + y^2 + z^2}} \mathbf{i} + \frac{x + 2y}{2\sqrt{x^2 + y^2 + z^2}} \mathbf{j} \implies \nabla f(3, -2) = \frac{2}{\sqrt{7}} \mathbf{i} - \frac{1}{2\sqrt{7}} \mathbf{j}.$$

5. The answer is (E).

Let $F(x, y, z) = xy^2z^3 - x^3yz^2 + 6$. Then $F(x, y, z) = 0$ gives $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{3x^2yz^2 - y^2z^3}{3xy^2z^2 - 2x^3yz}$.

Evaluating at $(x, y, z) = (1, 2, -1)$ we obtain $\frac{\partial z}{\partial x} = \frac{10}{16} = \frac{5}{8}$.

6. To use the formula $\kappa(x) = \frac{|y''|}{[1 + (y')^2]^{3/2}}$, we compute $y' = \frac{e^{-x}}{1 - e^{-x}} = \frac{1}{e^x - 1}$ and $y'' = \frac{-e^x}{(e^x - 1)^2}$.

At $x = \ln 2$ we obtain $y' = 1$ and $y'' = -2$. Thus $\kappa(\ln 2) = \frac{2}{[1 + (-1)^2]^{3/2}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$.

7. Since $\|\mathbf{a}\| = \sqrt{2}$, we use the unit vector $\mathbf{u} = \frac{1}{\sqrt{2}} \mathbf{a} = \frac{1}{\sqrt{2}} \mathbf{i} - \frac{1}{\sqrt{2}} \mathbf{j}$.

$$\begin{aligned} D_{\mathbf{u}}f(2, 1) &= \lim_{h \rightarrow 0} \frac{f\left(2 + \frac{h}{\sqrt{2}}, 1 - \frac{h}{\sqrt{2}}\right) - f(2, 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\left(2 + \frac{h}{\sqrt{2}}\right)^2 - \left(1 - \frac{h}{\sqrt{2}}\right)^2 - 3 \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[3 \left(1 + \frac{2h}{\sqrt{2}}\right) - 3 \right] = \lim_{h \rightarrow 0} \frac{1}{h} [3\sqrt{2}h] = 3\sqrt{2}. \end{aligned}$$

8. By the chain rule we have

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} = f_x(e^r \cos \theta) + f_y(e^r \sin \theta) \quad \text{and} \quad \frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta} = f_x(-e^r \sin \theta) + f_y(e^r \cos \theta).$$

Therefore,

$$\begin{aligned} \left(\frac{\partial w}{\partial r}\right)^2 + \left(\frac{\partial w}{\partial \theta}\right)^2 &= e^{2r} (f_x \cos \theta + f_y \sin \theta)^2 + e^{2r} (f_y \cos \theta - f_x \sin \theta)^2 \\ &= e^{2r} [(f_x)^2 \cos^2 \theta + 2f_x f_y \cos \theta \sin \theta + (f_y)^2 \sin^2 \theta \\ &\quad + (f_y)^2 \cos^2 \theta - 2f_y f_x \cos \theta \sin \theta + (f_x)^2 \sin^2 \theta] \\ &= e^{2r} [(f_x)^2 + (f_y)^2]. \end{aligned}$$

9. (a) The given surface is a level surface of the function f , thus the desired tangent plane is normal to $\nabla f(-1, 2, 1)$ and passes through the point $(-1, 2, 1)$. We have

$$\nabla f = \langle (y+z)e^{x+z}, e^{x+z}, (y+z+1)e^{x+z} \rangle \implies \nabla f(-1, 2, 1) = \langle 3, 1, 4 \rangle.$$

Thus an equation for the tangent plane is

$$3(x+1) + 1(y-2) + 4(z-1) = 0.$$

- (b) From part (a) we know

$$f_x(-1, 2, 1) = 3, \quad f_y(-1, 2, 1) = 1, \quad \text{and} \quad f_z(-1, 2, 1) = 4.$$

On the other hand, $f(-1, 2, 1) = 3$. Hence the linearization of f at $(-1, 2, 1)$ is

$$L(x, y, z) = 3 + 3(x+1) + 1(y-2) + 4(z-1).$$

Using this we obtain

$$f(-1.03, 2.01, 0.98) \approx L(-1.03, 2.01, 0.98) = 3 + 3(-0.03) + 1(0.01) + 4(-0.02) = 2.84.$$

10. First order partial derivative of f are

$$f_x = 2x(1-y) \quad \text{and} \quad f_y = 4y - x^2.$$

Since f_x and f_y exist for every (x, y) , the only critical points of f are solutions of the system

$$\begin{cases} 2x(1-y) = 0, \\ 4y - x^2 = 0. \end{cases}$$

The first equation gives $x = 0$ or $y = 1$. Substituting these in the second equation we obtain three critical points $(0, 0)$, $(-2, 1)$, and $(2, 1)$.

Now the second order partial derivatives of f are

$$f_{xx} = 2(1-y), \quad f_{xy} = f_{yx} = -2x, \quad \text{and} \quad f_{yy} = 4.$$

Hence the discriminant D is given by $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = 8(1-y) - 4x^2$.

Since $D(-2, 1) < 0$ and $D(2, 1) < 0$, it follows that f has saddle points at $(-2, 1)$ and $(2, 1)$.

On the other hand, since $D(0, 0) > 0$ and $f_{yy}(0, 0) > 0$, the function f has a local minimum at $(0, 0)$.